

OCTAVIA KEMETE - DATA ANALYTICS WITH SQL

SQL FUNDAMENTALS: SELECT & FILTERING

EXERCISE 01 - SQL FOUNDATIONS

01 SELECT *

FROM employees-table;

↳

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

02 SELECT DISTINCT (department)

FROM employees-table;

↳

department
IT
HR
Finance
Marketing

03 SELECT first-name,

last-name

FROM employees-table

ORDER BY salary, DESC;

↳

first-name	last name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4 SELECT salary

FROM employees-table
WHERE salary
LIMIT 3;

id	first-name	last-name	Salary
1	Alice	Green	10000
3	Cathy	Zulu	85000
5	Eva	Naidoo	72000

Q4 SELECT id,

first-name,

last-name,

salary

FROM employees-table

ORDER BY salary DESC

LIMIT 3;

↳

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cabny	Zulu	65 000

Q5 SELECT id,

first-name,

last-name,

department

FROM employees-table

WHERE department = 'IT';

↳

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6 SELECT *

FROM employees

WHERE department = 'finance' AND salary > 60000

↳

id	first-name	last-name	department	salary
3	Cabny	Zulu	finance	65000

07 SELECT DISTINCT (department)

FROM employees

WHERE department = 'HR' OR 'Marketing'

↳

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08 SELECT DISTINCT (department)

FROM employees

WHERE department NOT = 'IT'

↳

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09 SELECT DISTINCT (department)

FROM employees

WHERE department IN 'IT', 'HR' OR 'Finance'

↳

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 SELECT *

FROM employees

WHERE department = 'IT' salary > 65000 AND city = 'Johannesburg'

↳

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg